Chapter 02 Raw Textbook Content

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Textbook of Minimally Invasive Spine Surgery, Concepts and Surgical Techniques

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2. Overview of Minimally Invasive Spine Surgery

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2.1 Introduction

Minimally invasive spine surgery (MISS) has emerged over the past three decades as a paradigm shift in the surgical management of spinal disorders, aiming to achieve equivalent or superior clinical and radiographic outcomes compared with traditional open procedures while minimizing approach-related morbidity. The core philosophy of MISS is to accomplish the surgical objectives, whether decompression, stabilization, or deformity correction, through techniques that preserve paraspinal musculature, reduce iatrogenic

injury to soft tissues, and limit the physiological stress of surgery [1, 2].

The impetus for MISS development arose from early observations that extensive muscle dissection and retraction in open spine surgery contributed to postoperative pain, blood loss, prolonged hospitalization, and impaired functional recovery [3]. The introduction of microsurgical techniques in the 1970s and, subsequently, tubular retractor systems in the 1990s provided the technical foundation for smaller, targeted surgical corridors [4]. Over time, advancements in intraoperative imaging, navigation, robotics, and specialized instrumentation have expanded the applicability of MISS from isolated lumbar disc herniations to complex multilevel degenerative disease, trauma, infection, tumor, and adult spinal deformity correction [5–7].

A robust body of literature supports the clinical efficacy of MISS. Randomized and comparative cohort studies have demonstrated that procedures such as minimally invasive transforaminal lumbar interbody fusion (MI-TLIF) are associated with reduced intraoperative blood loss, lower postoperative pain scores, shorter hospital stays, and comparable fusion rates relative to open techniques [8, 9]. Similarly, minimally invasive approaches to adult spinal deformity have yielded meaningful improvements in sagittal and coronal alignment with lower perioperative morbidity in appropriately selected patients [10, 11]. Beyond patient-centered benefits, MISS offers potential health system advantages. Shorter length of hospital stay, earlier mobilization, and reduced complication